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Transformer-based models for sentiment analysis of YouTube video comments

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ABSTRACT

The rapid expansion of online video content has positioned platforms like YouTube as substantial sources of multimedia data, rich in diverse user interactions and expressions. Among these, comments have emerged as a significant source of sentiment, opinion, and feedback. The objective of this research is to improve comprehension of the emotions conveyed in YouTube comments through the assessment and comparison of transformer-based sentiment analysis algorithms. The main objectives were to assess various transformer architectures, including BERT, GPT, RoBERTa, and T5, for their effectiveness in sentiment classification tasks. The research utilized the Just Dance dataset for training and evaluated the models on several state-of-the-art sentiment analysis datasets to ensure robustness and generalizability. Methods involved exploring different pre-training tech- niques, loss functions, optimizers, and hyperparameter settings to optimize model performance. The findings reveal that RoBERTa consistently demonstrated superior performance, achieving the highest accuracy and F1-scores across various configurations, particularly with its specialized to- kenizer and the AdamW optimizer. This study contributes novel insights into the strengths and limitations of transformer-based approaches for analyzing YouTube comments and offers valu- able recommendations for improving sentiment analysis frameworks across diverse contexts while suggesting avenues for future improvements.

Introduction

Platforms like YouTube have become colossal repositories of multimedia data, facilitating di- verse forms of expression and interaction. Among the myriad ways users engage with this content, comments serve as a rich source of sentiment, opinions, and feedback. Understanding the senti- ments expressed in these comments provides not only insights into audience reactions but also aids content creators, marketers, and researchers in gauging the reception of their videos. However, manually analyzing the sentiment of a large volume of YouTube comments is a daunting task, prone to subjectivity and inefficiency [1].

Traditional sentiment analysis approaches, relying on rule-based methods or simplistic algo- rithms of machine learning, often struggle to capture the nuances of natural language and the contextual intricacies present in online discourse. Additional information about transformer fea- tures is provided in Appendix A.1. There is, therefore, a pressing need for more advanced and efficient techniques to automate the sentiment analysis of YouTube video comments [2].

Transformer-based models, introduced in 2017 by Vaswani et al. in their paper "Attention is All You Need" published at the NeurIPS

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conference, have revolutionized the natural language processing (NLP) field. Further theoretical background on transformer architectures is provided in Appendix A.1. This architecture has outperformed its predecessors, such as Con- volutional Neural Networks (CNNs) ans Recurrent Neural Networks (RNNs), due to its attention mechanism, which allows for parallel processing of sequential data, thus enhancing efficiency and the capacity to represent distant relationships in text sequences [3].

These transformer models, particularly the multi-headed attention mechanism, enable the model to weigh the importance of each word in a sentence relative to other words, thereby fa- cilitating the understanding of contextual relationships. Details on the attention mechanism and its variations across different models are explored in Appendix A.0.1. The basic struc- ture of a transformer comprises encoding and decoding blocks, where encoding transforms inputs into a series of contextual representations, and decoding generates outputs from these representa- tions. Additionally, transformers utilize positional embeddings to maintain the order of words in a sequence [4].

Transformer-based models have emerged as powerful tools for natural language understanding tasks, including sentiment analysis. Pre-trained on vast text corpora and fine-tuned for specific tasks, these models have shown outstanding results on several benchmarks for natural language processing [5].

Since the introduction of transformers, several landmark models have been developed, each bringing significant improvements to text comprehension and generation. Among these models, Bidirectional Encoder Representations from Transformers (BERT), released in 2018 by Google, has established new benchmarks due to its capability to capture complex contexts in both direc- tions. The Generative Pre-Trained Transformer (GPT) from OpenAI, with its text generation capabilities, offers contextually relevant responses. The BERT technique that is robustly tuned (RoBERTa), developed by Facebook AI, has taken BERT's performance even further, thanks to optimized training strategies and the use of large quantities of data. Finally, Google's Text-To- Text Transfer Transformer (T5) model unifies the approach to NLP tasks by processing them all in a text-to-text format, facilitating diverse applications such as translation, classification, and text generation [6].

Transformers are very useful since they can identify long-distance dependencies and contex- tual information through the self-attention mechanism, making them well-suited for analyzing the nuanced language in YouTube comments. Textual Sentiment Analysis, a major application of transformers in the field of natural language processing, has represented a significant advance in the ability to understand and interpret emotions and opinions expressed through written language. However, what is truly meant by "feeling" in this complex context? In the context of textual sen- timent analysis, sentiment encompasses much more than just raw emotional reactions. Rather, it refers to the essence of emotions and opinions encapsulated in textual data, the subtle nuances that shape our understanding of the world around us. The goal of sentiment analysis is to determine whether various types of text, such as comments, reviews, or social media posts, express a positive, negative, or neutral tone [7]. This sophisticated process requires careful deconstruction of words, phrases, and even the wider contexts in which these texts are produced and consumed. Thus, textual sentiment analysis is not limited to a simple classification of emotions but also involves a deep understanding of the social, cultural, and linguistic dynamics underpinning every act of written communication [8].

Beyond its use in social media and business, sentiment analysis also finds significant applica- tion in the field of mental health. The ability to detect and understand the feelings expressed in texts can be harnessed to monitor the emotional well-being of individuals on a large scale. For example, social networks have become spaces where people openly share their thoughts and emo- tions, offering a wealth of potentially valuable data for detecting early signs of emotional distress or mental disorders. Sentiment analysis techniques can be utilized to identify indicators of stress, depression, or anxiety, enabling early intervention and appropriate management. Additionally, this approach can contribute to a better understanding of the environmental and social factors influencing mental health, paving the way for more targeted and effective interventions and public health policies. In short, sentiment analysis offers promising potential for improving the early detection and management of mental health problems in modern society [9].

In this study, the principles are applied to a specific dataset, the Just Dance dataset. By analyzing the comments associated with videos from this dataset, the objective is to understand how transformer-based models perform sentiment analysis and reveal audience reactions. The study assesses the impact of Just Dance content on viewers, both positive and negative, through the analysis of YouTube comments. The use of the Just Dance as a dataset is explained by several pertinent reasons. Firstly, the comments associated with this type of content on YouTube are often rich in emotion and nuance, reflecting a variety of user reactions. This makes them an ideal testing ground for evaluating the capabilities of transformer-based models in sentiment analysis. Comments can oscillate between positive feelings, expressing joy and enthusiasm, and negative ones, such as frustration or disappointment, allowing us to explore the ability of models to detect and classify these varied emotions. In addition, the Just Dance dataset is extremely popular, attracting a wide range of viewers from different age groups and cultural backgrounds. This diversity in comments allows us to assess the robustness and generalizability of the models, by testing them on data that incorporates varied linguistic expressions, cultural ref- erences and distinct communication styles. By examining this dataset, the study not only analyzes the performance of the models in a specific context, but also contributes to a better understanding of the sentiments expressed in online interactions around video games.

Few studies have specifically focused on the comparative evaluation of transformer-based ar- chitectures in the context of YouTube comments. Existing research often centers on standardized datasets, such as the IMDB dataset for movie reviews or the Sentiment140 dataset for Twitter sen- timent analysis [10,11]. While these datasets provide valuable insights, they do not fully capture the unique linguistic and contextual nuances present in YouTube comments, such as colloquial language, emojis, and multimodal references. The lack of focused research limits the ability to effectively apply these models to YouTube's diverse and dynamic content [12]. Addi- tionally, most existing studies primarily evaluate transformers on general-purpose sentiment tasks, leaving a gap in domain-specific analyses tailored to platforms like YouTube [13].

This paper proposes a comparative study of transformer-based models for sentiment analysis of YouTube video comments, with a particular focus on the Just Dance dataset. Different trans- former architectures, pre-training techniques, and fine-tuning strategies will

be explored to assess their effectiveness in capturing sentiment nuances in this specific domain. Furthermore, the study examines the impact of factors such as comment length, user engagement, and video content type on the performance of these models. Through empirical evaluation and in-depth analysis, this study aims to provide insights into the strengths, limitations, and practical considerations of using transformer-based approaches for sentiment analysis in the context of YouTube video comments. The rise of online video-sharing platforms such as YouTube has generated an ocean of mul- timedia content, offering a wealth of data rich in user interaction and expression. Comments on these videos have become an essential vector of feelings, opinions, and feedback, revealing valuable insights into viewers' perceptions and behaviors. Over the years, numerous studies have explored the sentiments expressed in YouTube comments, utilizing a variety of natural language processing (NLP) techniques and machine learning models. However, despite significant advances in this field, there remain notable gaps in our understanding of the effectiveness of different transformer-based architectures for this specific task.

Transformer-based models, such as GPT, BERT, RoBERTa, and T5, have been widely adopted for sentiment classification tasks due to their ability to capture complex relationships in textual data. However, few studies have focused specifically on the comparative evaluation of these models in the context of YouTube comments. Moreover, most of the existing research has centered on standardized datasets, overlooking the specificity of interactions and sentiments expressed in online video environments. This lack of focused research limits practitioners' ability to effectively and appropriately apply these models to the analysis of YouTube comments.

This study evaluates and compares several transformer-based models for sentiment analysis in YouTube comments, utilizing the Just Dance dataset for training. The assessment extends to other state-of-the-art sentiment analysis datasets, ensuring robustness and generalizability of the results. The research objectives focus on exploring various pre-training techniques, loss functions, optimiz- ers, and hyperparameter settings to enhance model performance. Preliminary results demonstrate that RoBERTa outperforms other models, achieving high accuracy and F1-scores, particularly due to its specialized tokenizer and the AdamW optimizer.

This study aims to contribute to the understanding of the strengths and limitations of transformer- based approaches to YouTube comment analysis, and to provide valuable recommendations for improving sentiment analysis frameworks in various contexts. Moreover, suggestions for future im- provements are provided, with the aim of guiding further research toward more tailored approaches to the specific challenges posed by online interactions.

Literature review

Natural Language Processing (NLP) is a domain within artificial intelligence that concentrates on enabling computers to understand, interpret, and interact with human language. It encom- passes a wide range of tasks, including sentiment analysis, machine translation, text generation, and speech recognition. Sentiment analysis, specifically, focuses on determining the sentiment or emotional tone of a text, classifying it as neutral, negative, or positive. This method is commonly employed in domains like marketing, customer support, and political studies to comprehend the viewpoints and feelings conveyed in textual data [14].

Sentiment analysis enables companies and organizations to gather valuable insights from large quantities of textual data, including comments on social networks, product reviews, and customer feedback. Thanks to this analysis, it is possible to detect trends, identify recurring problems, and make informed decisions to improve products and services.

However, sentiment analysis and natural language processing pose several challenges. Contex- tual understanding, linguistic ambiguities, and dialect variations are obstacles to overcome. In addition, the speed and efficiency of NLP models are crucial for processing large amounts of data in real time. These challenges have led to major developments in the field of NLP models, moving from traditional techniques to advanced architectures such as transformers [15].

Transformers, developed by Vaswani et al. in 2017, have revolutionized the field of NLP by proposing a new approach based on the attention mechanism. This approach enables models to focus on the relevant parts of the text, improving processing accuracy and efficiency. Since their introduction, transformers have been the basis of many successful models, such as BERT and GPT, which have shown impressive results in various NLP tasks, including sentiment analysis [16].

The aim of this literature review has been to provide an in-depth examination of recent advances in the use of transformers for text-based sentiment analysis. The evolution of NLP models, the characteristics of transformers, the main models based on this architecture, and their specific applications to sentiment analysis have been explored. In addition, current challenges and future prospects in this rapidly expanding field have been discussed

Evolution of NLP models

The evolution of natural language processing (NLP) models has gone through several significant phases, each bringing significant advances in terms of performance and language comprehension capabilities. Early NLP approaches relied mainly on statistical methods and linguistic rules. Tech- niques based on n-gram language models, syntactic trees and hidden Markov networks (HMMs) were commonly used for tasks such as speech recognition and machine translation. These meth- ods used text corpora to estimate word sequence probabilities but were limited by their ability to capture long-term dependencies in text. For example, n-gram models cannot effectively handle semantic relationships between words beyond a fixed context of n words [17].

Recurrent neural networks (RNN)

The emergence of artificial neural networks paved the way for Recurrent Neural Networks (RNNs), a pivotal advancement in NLP. RNN variants, such as Long Short-Term Memory (LSTM) and Gated Recurrent Units (GRU), excelled at capturing temporal dependencies and contextual information within sequential data. LSTMs, for example, use gating mechanisms to control the flow of

information, making it possible to manage long-term dependencies and alleviate the problem of vanishing gradient. This has significantly improved performance in tasks such as machine translation, text generation and speech recognition. However, despite these improvements, RNNs remained limited by their ability to process very long sequences due to the sequential nature of their processing, resulting in high computation time [18].

Word embeddings

Another major advance was the introduction of word embeddings, such as Word2Vec and GloVe, which enabled words to be represented as continuous vectors in high-dimensional space. These vector representations captured the semantic similarities between words, improving the performance of NLP models. For example, Word2Vec uses skip-gram and continuous bag-of- words (CBOW) architectures to learn context-based representations of words in large text corpora. This has facilitated the use of neural networks for more complex tasks, providing a solid basis for the representation of textual data. Word embeddings have enabled better generalization and a reduction in the dimensionality of textual data, making models more efficient and accurate [19].

Comparative study of transformer-based models

Transformer models, such as BERT, GPT, RoBERTa, and T5, each have their own unique characteristics and distinct performance in different natural language processing (NLP) tasks. A detailed comparative study of these models helps us to understand their respective strengths and weaknesses. Bidirectional Encoder Representations from Transformers (BERT): BERT is designed to understand the context of words by considering both their left and right surroundings within a sequence, using a bidirectional approach. This allows the model to capture deeper contextual relationships compared to unidirectional models. BERT is pre-trained on two main tasks: Masked Language Modeling (MLM) and Next Sentence Prediction (NSP). In MLM, certain words in a sentence are masked, and the model is tasked with predicting these missing words, enhancing its understanding of context. BERT excels in natural language comprehension tasks such as text classification, question-answering, and sentiment analysis. However, due to its bidirectional nature, BERT is less suited for text generation tasks.

Generative Pre-trained Transformer (GPT): GPT employs a unidirectional encoder architec- ture, where each word in a sequence depends only on the preceding words. This design is partic- ularly well-suited for text generation, as it allows for smooth and coherent sentence production. The model is pre-trained on a vast amount of text using an unsupervised language modeling task, which enables it to generate text that is contextually relevant and coherent. While GPT excels in generating text, it is less effective in tasks that require a two-way understanding of context, such as those necessitating comprehension of both preceding and following text [20].

Robustly Optimized BERT Approach (RoBERTa): RoBERTa is an optimized version of BERT that enhances performance by fine-tuning hyperparameters, expanding the pre-training corpus, and removing the next-sentence prediction task. It undergoes intensive training on larger and more diverse datasets for longer durations, which results in a more robust model. As a result, RoBERTa achieves significantly improved performance on various NLP benchmarks compared to BERT. However, this optimization comes at the cost of higher computational resource requirements for training. RoBERTa is used in similar applications as BERT but with enhanced performance [21]. Text-To-Text Transfer Transformer (T5): T5 uses a unified text-to-text framework for all NLP tasks, where both input and output are represented as text strings. This approach allows T5 to handle a variety of NLP tasks within a single model architecture. It is pre-trained on a diverse set of tasks using a variant of masked language modeling, enabling the model to be adaptable to various applications. T5's flexibility allows it to excel in tasks such as machine translation, text summarization, and classification. However, its configuration can be more complex for specific tasks compared to specialized models [22].

Table 1

Transformers for sentiment analysis

The application of transformers to sentiment analysis marks a significant turning point in the field of natural language processing (NLP). These models have significantly improved accuracy and the ability to capture the fine-grained semantics of language, essential

Table 1
Comparison of transformer models.

Model	Architecture	Strengths	Weaknesses	Key Applications
BERT (Bidirectional Encoder Representations from Transformers)	Bidirectional	Deep contextual understanding	Limited text generation capability	Text classification, Question Answering (QA), Named Entity Recognition (NER)
GPT (Generative Pre-trained Transformer)	Unidirectional	Fluent and coherent text generation	Less effective in contextual understanding	Text generation, Conversational AI
RoBERTa (Robustly Optimized BERT Approach)	Optimized Bidirectional	Improved NLP performance	High computational cost	General NLP tasks with enhanced performance
T5 (Text-To-Text Transfer Transformer)	Text-to-Text	High flexibility for various NLP tasks	Complex configuration	Translation, Text summarization, Classification

for interpreting the nuances of human sentiment expressed in text.

Current model training methodologies and data pre-processing

Transformer-based models for sentiment analysis adopt sophisticated training methodologies. Initially pre-trained on large unannotated text corpora, such as the BooksCorpus and Wikipedia, these models are then fine-tuned on specific sentiment analysis tasks using annotated datasets such as IMDB for positive and negative sentiment, and SST-2 for binary sentiment. Before being fed into the model, textual data is cleaned to eliminate noise, such as HTML tags, special characters, and redundant punctuation. Next, tokenization divides sentences into tokens, the basic units understood by the model, while adding special tokens to indicate the beginning and end of the text. This prepares the text data to be compatible with the input requirements of the transformers, ensuring an accurate and structured representation of the text for sentiment analysis [23].

Transformers have been successfully applied to a variety of use cases in sentiment analysis. For example, BERT has demonstrated an exceptional ability to capture the contextualization of language, improving the accuracy of sentiment predictions over previous models. Comparative studies have shown that variants such as RoBERTa, by optimizing BERT's training approach, often outperform its initial performance in specific benchmarks such as sentiment classification tasks. In addition, real-life applications in fields such as social media monitoring, customer feedback analysis, and political opinion assessment have illustrated the adaptability and effectiveness of transformers in a variety of contexts [24].

Challenges and limitations of transformers in sentiment analysis

Transformers have revolutionized sentiment analysis, yet they come with several challenges and limitations that need to be addressed for broader application. One major issue is the complexity and computational cost associated with these models. Transformers, especially large models like BERT and GPT-3, can have billions of parameters, requiring significant computational resources for training and inference. For instance, training a model like GPT-3 can incur costs of several million dollars due to the high computational power needed. This makes it difficult to deploy these models in resource-limited environments and poses challenges for large-scale optimization and deployment [25].

Another critical concern is data and model bias problems. Since transformer models are often pre-trained on vast corpora from the Internet, they can inadvertently learn and perpetuate biases present in these data sources. This can lead to the models exhibiting cultural stereotypes, racial biases, sexism, and other prejudices. For example, a transformer model might reflect certain political or social biases if the training data predominantly originates from sources with these inclinations. Such biases raise ethical concerns and can compromise the reliability of sentiment analyses, especially in sensitive applications [26].

Additionally, generalization and learning transfer remain significant challenges. While trans- formers are adept at learning from extensive data, they often struggle to generalize across different contexts or domains. A model fine-tuned on one dataset may not perform well in another do- main without additional adaptation. For example, a model trained on movie reviews may fail to accurately analyze electronics reviews without further fine-tuning. This limitation hampers the universal applicability of transformer models and underscores the need for ongoing research to enhance their generalization and transfer learning capabilities [27].

Future directions in transformer research

To overcome the challenges of transformers, various research directions are being explored. This section discusses promising avenues for improving models and applying them to sentiment analysis. Optimizing transformer models for specific tasks is a key research direction. This includes the development of smaller, more efficient models that retain the performance of larger models while reducing computational requirements. Techniques such as model distillation, where a large "teacher" model transfers its knowledge to a smaller "student" model, show promise in achieving this goal. In addition, the creation of specialized models for particular domains, such as finance or healthcare, can improve the accuracy and relevance of sentiment analyses [28].

Improving class pre-processing and balancing techniques continues to be an active area of research. New tokenization methods, which take better account of linguistic and contextual sub- tleties, can improve the quality of input data. In addition, advanced class balancing techniques, such as the use of synthetic data generators or intelligent resampling algorithms, can help overcome class imbalance problems, leading to more robust and fair models. The exploration of new hybrid architectures, combining the advantages of transformers with other machine learning techniques, offers promising prospects. For example, integrating convolutional neural networks (CNNs) to ex- tract local features with the global capabilities of transformers can improve performance. Similarly, the use of ensemble models, where several models are combined to produce more robust predic- tions, can also lead to significant improvements. These hybrid approaches aim to leverage the complementary strengths of different methods to overcome the limitations of transformers alone [29].

Summary and synthesis

In summary, this literature review explored in depth the progression of natural language pro- cessing (NLP) models, emphasizing the shift from recurrent neural networks (RNNs) and long short-term memory networks (LSTMs) to transformer architectures. This transition marked a crucial milestone in the field, with models such as BERT, GPT, RoBERTa and T5 demonstrat- ing significant advances in sentiment analysis [30]. Transformers have transformed the way NLP models process and understand textual data, thanks to their attention mechanism, which captures long-term dependencies in text. Additional information on the attention mechanism discussed in Appendix A.0.1. These models have demonstrated an exceptional ability to capture the complexity and nuances of human language, which is crucial for accurate sentiment analysis.

However, despite their impressive performance, transformers are not without their challenges. Their complexity and high computational cost, inherent biases in training data, and difficulties in generalizing to specific domains are all obstacles that require ongoing attention. Researchers are actively working on solutions to optimize these models, improve preprocessing and class balancing techniques, and explore new hybrid architectures that combine the strengths of transformers with other approaches. Future research avenues are promising and varied. Optimizing models for specific tasks, developing more advanced preprocessing techniques, and exploring new hybrid architectures are all avenues that could overcome current limitations and open up new perspectives for sentiment analysis and other NLP applications.

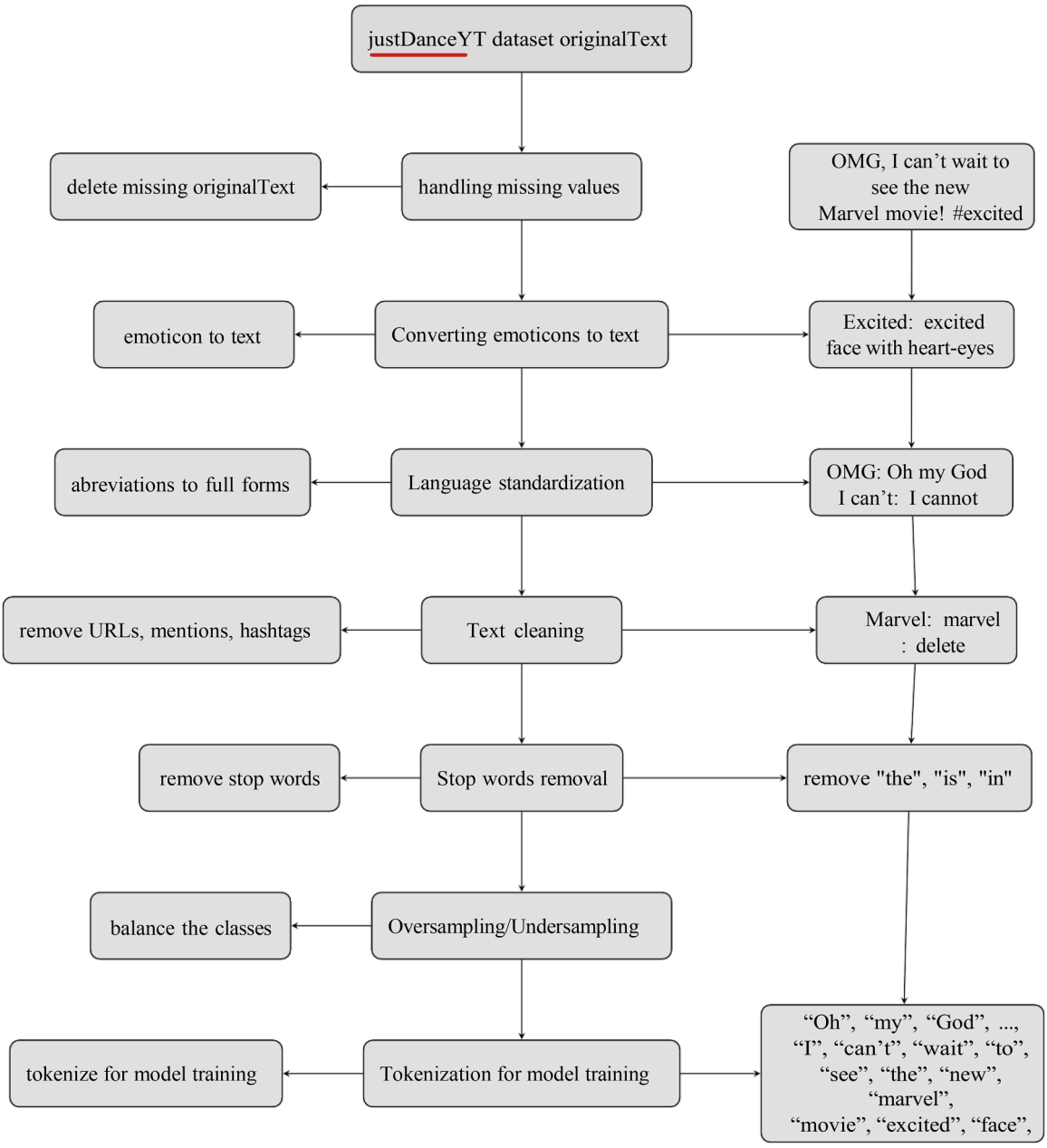


Fig. 1. Text preprocessing steps diagram.

Methodology

Key research questions

This article addresses several research questions regarding the optimization and performance of sentiment analysis models. Specifically, it seeks to answer the following:

- How do different tokenizers, loss functions and optimization algorithms (Adam, SGD, RM- Sprop…) impact the performance of transformer models for sentiment analysis?
- How do BERT, GPT, RoBERTa, and T5 perform under optimal conditions across accuracy, precision, recall, and F1-score metrics in sentiment classification?
- What is the optimal learning rate, batch size, and other key training hyperparameters for achieving optimal performance with these transformer models in sentiment analysis?
- How do the optimized models compare in terms of strengths and weaknesses on various sentiment analysis datasets (IMDb, SST-2, Yelp, Amazon Reviews)?
- What are the findings of this study and its limitations?
- What recommendations can be made for improving transformer-based sentiment analysis models and for future research directions?

Data preparation

In this section, the data preparation process for the comprehensive analysis utilizing the Just Dance dataset is presented. This dataset served as the cornerstone for our investigation into viewer engagement, sentiment, and broader interaction patterns. This rich repository encom- passes 278,242 YouTube comments and video metrics across 16 distinct variables. The key vari- ables include unique identifiers for comments (commentID), comment text (originalText) for sen- timent and theme analysis, pre-computed sentiment scores (sentiment), number of likes (likes), comment posting date (dateComment), comment type (mainComment), channel identifiers and titles (channelID, channelTitle), video identifiers and titles (videoID, videoTitle), video posting date (dateVideo), engagement metrics (viewsVideo, likesVideo, dislikesVideo), total comments per video (total-CommentsVideo), and video descriptions (descriptionVideo). These variables enable detailed analysis of viewer engagement, sentiment distribution, and interactions across different videos and channels.

To effectively manage data preprocessing, where data irregularities such as missing values, outliers, and inconsistencies are prevalent, this section details the data preprocessing process em- ployed in this study. It is a critical initial step in the development of deep learning models like transformers. As depicted in Fig. 1, the process encompasses many steps:

- Load the ‘justDanceYT ‘dataset within which each comment is characterized by an ‘original- Text ‘field encapsulating the raw textual content. This foundational step involves importing the raw data necessary for the subsequent stages of the preprocessing pipeline.
- Missing value management is the first step after data loading. It involves identifying and deleting comments that have no text (‘originalText ‘is missing). Machine learning models require complete and consistent inputs. Removing missing data prevents errors and improves dataset quality [31].
- The conversion of emoticons into text follows the management of missing values. This step transforms emoticons into their text equivalents. For example, an emoticon of a smiling face becomes "smiling face". Language models handle words better than symbols. This conversion helps the models to understand the feelings expressed by the emoticons [32].

For the transformation of emoticons into text, a specific approach was adopted to convert emo- tional symbols into their textual equivalents. For example, an emoticon representing a smiling face was converted to "smiling face," and other emoticons were similarly transformed. This conversion was performed using Python libraries such emoji and re (regular expressions), which allow for the identification of emoticons and their efficient replacement with corresponding textual representa- tions. This step is crucial as language models process words better than symbols, enhancing their ability to understand the emotions and sentiments expressed by users in comments.

- The language is then standardized. This involves replacing abbreviations with their full forms, e.g. "OMG" becomes "Oh my God" and "can’t" becomes "cannot". This step ensures linguistic consistency, enabling models to better understand and learn from data.
- Text cleansing is a crucial step where URLs, mentions, hashtags and other non-text elements are removed. For example, "https://", "@user" and "topic" are eliminated. This step eliminates distractions and noise, making the text cleaner and more relevant for sentiment analysis.
- The next step is to remove stop words. Stop words, such as "the", "is", "in", are frequent words that don’t carry a strong meaning and can add noise to the data. For example, in the sentence "I can’t wait to see the new Marvel movie!", words like "the" and "to" would be removed. Deleting these words reduces noise and focuses the analysis on meaningful words, thus improving model performance.

- After cleaning and stop words, the text is transformed into a list of tokens (words or sub- words) by tokenization. For example, "I can't wait to see the new Marvel movie!" becomes ['I', 'cannot', 'wait', 'to', 'see', 'the', 'new', 'Marvel', 'movie']. This step prepares the data for con- sumption by transformer models, which process sequences of tokens.
- To manage class imbalances between positive, neutral and negative sentiments, a data re- balancing step is included. Rebalancing can be done by oversampling, artificially increasing the number of neutral and negative comments by duplicating certain examples, or by undersampling, reducing the number of positive comments by random sampling. Class imbalances can bias ma- chine learning models, making them less effective for minority classes. Rebalancing ensures that the model is fairly trained on all sentiment classes.
- Finally, the prepared data is used to train the transformer models. The models learn to predict the sentiments of new comments based on patterns identified in the training data. This final phase is crucial, as the quality of the training depends heavily on the quality of the data preparation [33].
- The meticulously preprocessed dataset serves as the foundation for the subsequent develop- ment and evaluation of deep learning models. The following section will delve into the architecture and training of the transformer models employed in this study, followed by a comprehensive eval- uation of their performance [34].

Model evaluation

This section outlines the evaluation procedure used for the studied transformer models. Com- parative results of the performance of the different studied transformer models will be compiled to present the performance of Generative Pre-trained Transformer (GPT), Bidirectional Encoder Representations from Transformers (BERT), Robustly optimized BERT approach (RoBERTa), and Text-To-Text Transfer Transformer (T5) on different datasets, in terms of precision, recall and F1-score. Then, a comparative analysis is conducted across multiple datasets, using common measures like F1-score, recall, and precision. These metrics provide a quantitative measure of model performance in terms of correctly classifying positive and negative sentiments [35].

To further optimize model performance, an in-depth exploration of tokenizers, loss functions, optimizers, and hyperparameters is undertaken. Different tokenization strategies will be compared to assess their impact on model performance. Numerous loss functions, such as focused loss and cross-entropy, will be evaluated to identify the most suitable approach for sentiment classification. Additionally, the impact of different optimizers, such as Adam, SGD, RMSprop, and AdamW, on model convergence and performance will be investigated. Hyperparameters, including learning rate, batch size, and number of layers, will be tuned to determine their influence on model accuracy. The results of these experiments will be presented through visualizations and detailed discussions to provide insights into optimal model configurations.

By combining these evaluation methodologies, a comprehensive understanding of model per- formance and the impact of different hyperparameters is achieved. This knowledge is essential for selecting the most effective model and configuration for sentiment classification tasks.

Experimentation

The effectiveness of transformer models in text-based sentiment analysis is compared through experimentation. Specifically, several transformer architectures, including BERT, GPT, RoBERTa, and T5, are focused on, and their performance on different sentiment analysis tasks is evaluated. The impact of various preprocessing techniques, loss functions, and optimizers on the perfor- mance of these models is also examined. Additionally, these models are applied to several well-known datasets in scientific research to obtain a comprehensive comparative evaluation.

Experimental setup

In this subsection, the experimental setup used to evaluate the transformer models for sentiment analysis is outlined. This includes evaluation metrics, loss functions, optimizers, and hyperparam- eter configurations. The goal is to provide a clear and replicable methodology for assessing the performance of the models under various conditions. By standardizing the experimental setup, con- sistency in the evaluation process is ensured, facilitating a fair comparison of results across different model configurations and techniques. The performance in sentiment analysis is rigor- ously evaluated using a training environment composed of high-performance GPU environment (NVIDIA Tesla V100), TensorFlow and PyTorch libraries. The definitions and roles of the pri- mary components of these configurations are detailed in the following subsections.

Evaluation metrics

To evaluate model performance on sentiment analysis tasks, several metrics are employed. The following metrics are calculated for each class C_i (negative, neutral, positive), denoted by index i where:

- K is the total number of classes.
- TP_i is the number of True Positives for class C_i.
- TN_i is the number of True Negatives for class C_i.
- FP_i is the number of False Positives for class C_i.
- FN_i is the number of False Negatives for class C_i.

- N is the total number of samples.

Accuracy: defines the proportion of correct predictions out of all predictions made. It is a global measure of model performance, indicating the frequency with which the model makes correct predictions. It can be expressed as:

$$\text{Accuracy} = \frac{\sum_{i=1}^{K} \text{TP}_\text{i} + \sum_{i=1}^{K} \text{TN}_\text{i}}{\text{N}} \tag{1}$$

Precision: evaluates the proportion of correct predictions for a given class among all predictions for that class. It is calculated for each class i as:

$$\text{Precision}_i = \frac{TP_i}{TP_i + FP_i} \tag{2}$$

Recall: evaluates the proportion of correct positive predictions among all the samples that are actually positive. This measure is crucial when the cost of false negatives is high. It is calculated for each class i as:

$$\text{Recall}_i = \frac{TP_i}{TP_i + FN_i} \tag{3}$$

The F1-Score provides a balance between the precision and recall metrics by evaluating the har- monic mean of the two. When assessing models on unbalanced datasets, it is helpful. The com- putation is done for every class i as:

$$\text{F1} - \text{Score}_i = 2 \times \frac{\text{Precision}_i \times \text{Recall}_i}{\text{Precision}_i + \text{Recall}_i} \tag{4}$$

Accuracy provides a quick and intuitive measure of model performance, offering an all-encompassing view; however, it can be misleading on unbalanced datasets where one class dominates. Precision becomes particularly critical in scenarios where false positives carry significant negative conse- quences, such as in sentiment analysis aimed at ensuring the correctness of positive or negative predictions for purposes like filtering comments for promotions or moderation. Recall, on the other hand, is essential in situations where missing a class incurs a high cost; for instance, capturing as many positive or negative samples as possible is vital in detecting reviews that demand immediate action. F1-Score serves as a balanced measure that harmonizes precision and recall, especially in unbalanced datasets, offering a comprehensive evaluation of model performance and becoming particularly relevant in sentiment analysis where both false positives and false negatives must be minimized.

Loss methods

Loss functions play a crucial role in model training as they quantify the difference between the training data's actual values and the model's predictions. The choice of an appropriate loss function can enhance overall model performance and influence its ability to generalize to unseen data. Several commonly used loss methods were examined, each characterized by distinct features impacting model convergence, the ability to handle class imbalance, and robustness to noise and atypical examples. These methods include cross-entropy loss, focal loss, hinge loss, Dice Loss, mean squared error (MSE), and Huber loss. Loss functions play a crucial role in model training as they quantify the difference between the training data's actual values and the model's predictions. The choice of an appropriate loss function can enhance overall model performance and influence its ability to generalize to unseen data. Several commonly used loss methods were examined, each characterized by distinct features impacting model convergence, the ability to handle class imbalance, and robustness to noise and atypical examples. These methods include cross-entropy loss, focal loss, hinge loss, Dice Loss, mean squared error (MSE), and Huber loss]èç.

Common Parameters:

- N is the total number of samples in the dataset.
- K is the total number of classes.
- $\hat{y_{ij}}$ is the predicted probability or value for the i th sample and j-th class.
- y_{ij} is the true value or indicator for the i th sample and j-th class.

Cross-entropy loss is used for classification problems and measures the divergence between the predicted probability distribution $\hat{y}$ and the actual label distribution y. The goal is to minimize this divergence to improve prediction accuracy. It can be expressed as:

$$\text{CrossEntropyLoss} = -\sum_{i=1}^{N} \sum_{j=1}^{K} y_{ij} \log\left(\hat{y}_{ij}\right) \tag{6}$$

Focal loss enhances cross-entropy loss by assigning more weight to hard-to-classify examples, which helps handle class imbalance better. It can be expressed as:

$$\text{FocalLoss}(\hat{y}, y) = -\sum_{i=1}^{N} \sum_{j=1}^{K} \alpha_j \left(1 - \hat{y}_{ij}\right)^{\gamma} y_{ij} \log\left(\hat{y}_{ij}\right) \tag{7}$$

- αj is a weighting factor for class j.
- γ is a focusing parameter that modifies how simple instances are down-weighted.

Hinge loss is used for training classifiers, especially for support vector machines (SVMs), and aims to maximize the margin between classes. It can be expressed as:

$\text{HingeLoss}(y, \widehat{y}) = \frac{1}{N} \sum_{i=1}^{N} \sum_{j \neq y_i} \max\left(0, 1 + \widehat{y}_{ij} - \widehat{y}_{iy_i}\right)$ (8)yi is the true class label for the i th sample.

- $\hat{y}_{iy}i$ is the predicted score for the correct class y_i.
- $\hat{y}_{ij}$ is the predicted score for class j.

Dice Loss measures the similarity between two sets, emphasizing intersection and union, and is useful for tasks where precise localization is important. It can be expressed as:

$\text{DiceLoss}(\widehat{y}, y) = 1 - \frac{2\sum_i \widehat{y}_i y_i}{\sum_i \widehat{y}_i^2 + \sum_i y_i^2}$ (9) i indexes the elements in the predicted and true label arrays.

- $\hat{y}_i$ and y_i are the predicted and true values for the i th element, respectively.The average squared error between the true and projected values is measured by the Mean Squared Error (MSE), which is frequently used in regression. It can be expressed as:

$$\text{MSE}(\widehat{y}, y) = \frac{1}{n} \sum_{i=1}^{n} \sum_{j=1}^{K} \left(\widehat{y}_{ij} - y_{ij}\right)^2 \tag{10}$$

- n is the total number of samples.because it is less susceptible to outliers and strikes a compromise between quadratic and linear loss, Huber loss combines the benefits of Mean Absolute Error (MAE) and Mean Squared Error (MSE). It can be expressed as:

$$\text{HuberLoss}(\widehat{y}, y) = \frac{1}{n} \sum_{i=1}^{n} \sum_{j=1}^{K} \begin{cases} \frac{1}{2}\left(\widehat{y}_{ij} - y_{ij}\right)^2 & \text{if } \left|\widehat{y}_{ij} - y_{ij}\right| \leq \delta \\ \delta\left(\left|\widehat{y}_{ij} - y_{ij}\right| - \frac{1}{2}\delta\right) & \text{otherwise} \end{cases} \tag{11}$$

- δ is a threshold for transitioning between quadratic and linear loss.

Optimizers

Optimizers play a crucial role in training neural networks by updating the weights of the model to minimize the loss function [1]. Different optimizers were designed to use various techniques to achieve this, and their performance significantly impacted the convergence speed and final accuracy of the model. In this section, several widely used optimizers were compared:

Common Parameters:

- θ: Model parameters (weights)
- η: Learning rate
- g_t: Gradient of the loss function at time step t
- ϵ: Small constant to avoid division by zero
- β_1, β_2: Exponential decay rates for moment estimates
- t: Time step
- $J(\theta)$: Loss function

Stochastic Gradient Descent (SGD): Uses a constant learning rate for each update.

$$\theta = \theta - \eta \nabla \text{J}(\theta) \tag{12}$$

where $\nabla J(\theta)$ is the gradient of the loss function with respect to the parameters θ. SGD often requires fine-tuning of hyperparameters for optimum performance.

Adam (Adaptive Moment Estimation): Utilizes first and second moment gradient esti- mates to adjust learning rates.

$$
\begin{aligned}
m_t &= \beta_1 m_{t-1} + (1-\beta_1) g_t \\
v_t &= \beta_2 v_{t-1} + (1-\beta_2) g_t^2 \\
\widehat{m}_t &= \frac{m_t}{1-\beta_1^t} \\
\widehat{v}_t &= \frac{v_t}{1-\beta_2^t} \\
\theta &= \theta - \eta \frac{\widehat{m}_t}{\sqrt{\widehat{v}_t} + \epsilon}
\end{aligned} \tag{13}
$$

where:

- m_t: First moment estimate (moving average of gradients)
- v_t: Second moment estimate (moving average of squared gradients)
- $\hat{m}_t$: Bias-corrected first moment estimate
- $\hat{v}_t$: Bias-corrected second moment estimate

Adam can dynamically adapt learning rates for each parameter, leading to faster and more stable convergence.

RMSprop (Root Mean Square Propagation): Adapts the learning rate according to the moving average of the squares of the gradients.

$$
\begin{aligned}
E\left[g^2\right]_t &= \beta E\left[g^2\right]_{t-1} + (1-\beta) g_t^2 \\
\theta &= \theta - \frac{\eta}{\sqrt{E[g^2]_t + \epsilon}} g_t
\end{aligned} \tag{14}
$$

where E[g2]t is the exponential moving average of squared gradients. RMSprop performed well, although competitive, it is slightly inferior to Adam and AdamW.

Adagrad (Adaptive Gradient Algorithm): Adapts learning rates according to the cumu- lative sum of the squares of the gradients.

$$
\theta = \theta - \frac{\eta}{\sqrt{G_{t,ii}} + \epsilon} g_t \tag{15}
$$

where Gt,ii is the accumulated sum of squared gradients for parameter iii up to time t. Adagrad can excessively reduce learning rates, slowing down learning over the long term.

AdamW (Adam with Weight Decay): Similar to Adam but with explicit weight regular- ization (weight decay).

$$
\begin{aligned}
m_t &= \beta_1 m_{t-1} + (1-\beta_1) g_t \\
v_t &= \beta_2 v_{t-1} + (1-\beta_2) g_t^2 \\
\widehat{m}_t &= \frac{m_t}{1-\beta_1^t} \\
\widehat{v}_t &= \frac{v_t}{1-\beta_2^t} \\
\theta &= \theta - \eta \left(\frac{\widehat{m}_t}{\sqrt{\widehat{v}_t} + \epsilon} + \lambda \theta \right)
\end{aligned} \tag{16}
$$

where λ is the weight decay coefficient for regularization.

Hyper-parameters

Finally, in this subsection, the hyperparameters to be examined, such as learning rate, number of layers, batch size, and other key parameters influencing language model training and perfor- mance, are explored. These hyperparameters are adjustable parameters defined before the start of training that directly impact the model learning process. They play a crucial role in training convergence, result quality, and learning speed. An overview of the main hyperparameters studied in this analysis is provided below:

- Learning Rate: Determines the size of the steps taken when optimizing model weights during training.
- Batch Size: Number of training examples used in a single optimization iteration.
- Number of Layers: Number of transformation layers in the model architecture.
- Epochs: Number of complete runs over the entire training dataset. A higher number of epochs allows the model to better learn the characteristics of the data, up to a point where overfitting may occur.
- Batch Train: Size of the batch of data used for training at each iteration. A larger batch size can stabilize training but increase memory consumption.

- Batch Eval: Size of data batch used for model evaluation at each iteration. Like the training batch size, this affects the speed and stability of the evaluation.
- Warmup Steps: Number of iterations during which the learning rate increases linearly from zero to the specified initial value. This helps stabilize initial training by avoiding large weight adjustments.
- Weight Decay: Regularization factor that penalizes large weights to avoid overfitting. This helps to keep model weights small and improve generalization.

Experimental results

This section presents a detailed analysis of the experimental results derived from applying trans- former models, specifically BERT, GPT, RoBERTa, and T5, to the task of sentiment classification. Various configurations were explored to identify optimal settings, including tokenization methods, loss functions, optimizers, and hyperparameter tuning. Performance was evaluated across multiple metrics, such as accuracy, precision, recall, and F1-score.

The choice of tokenization method emerged as a critical factor influencing model performance. Both model-specific tokenizers, such as BertWordPiece for BERT and RoBERTa's dedicated tok- enizer, and generic methods, such as SentencePiece and Byte-Pair Encoding (BPE), were assessed. Model-specific tokenizers consistently provided superior results, reflecting their alignment with the respective architectures. For instance, RoBERTa achieved its best F1-score of 0.93 with its dedi- cated tokenizer, while SentencePiece delivered competitive but slightly lower performance across all models. In contrast, generic tokenizers, such as Word Tokenizer, failed to match the effectiveness of model-specific approaches, as seen in GPT, where the F1-score dropped to 0.75. These find- ings emphasize the necessity of selecting tokenization methods tailored to each model to achieve optimal results.

The selection of loss functions also played a vital role in determining the quality of model predictions. Various loss functions were evaluated, including Cross-Entropy, Focal Loss, Hinge Loss, Dice Loss, Mean Squared Error (MSE), and Huber Loss. Among these, Cross-Entropy emerged as the most reliable across all transformer models, yielding consistently high precision, recall, and F1-scores. For example, RoBERTa achieved an F1-score of 0.93 using Cross-Entropy, compared to 0.88 with Focal Loss. Models like GPT and T5 showed significant sensitivity to the choice of loss function, with suboptimal scores when trained with Hinge Loss or Dice Loss. This observation underscores the general compatibility of Cross-Entropy with transformer architectures for sentiment classification. in terms of optimization, five popular algorithms were evaluated: SGD, Adam, RMSprop, Ada- grad, and AdamW. AdamW consistently outperformed the others, enabling RoBERTa to achieve its highest F1-score of 0.93. In comparison, SGD exhibited inferior performance across all mod- els, with an F1-score of only 0.74 for GPT. These findings indicate that optimizers with adaptive learning rates, like Adam and AdamW, are particularly suited to fine-tuning transformer models. Hyperparameter tuning was another focus of this study, with specific attention given to learn- ing rate, batch size, epochs, warmup steps, weight decay, and evaluation strategy. The optimal configuration for all models included a learning rate of 3e-5, a batch size of 32, and 500 warmup steps. Weight decay of 0.01 was found to enhance generalization, while evaluation at regular steps provided more granular control over performance monitoring compared to epoch-based evaluation.

Table 1 encapsulates the experimental outcomes and hyperparameter configurations employed for transformer models in senti- ment analysis. It comprehensively displays critical information, including the models' tokenizers, optimizers, loss functions, and associated hyperparameters such as learning rate, batch size, and number of epochs. Additionally, the table highlights key metrics, namely accuracy, precision, recall, and F1-score, which are strategically placed at the end for enhanced clarity. These results also underscore the superior choices for tokenizers, optimizers, and loss functions for each model, highlighting their effectiveness in driving performance improvements.

Table 2

In summary, the findings reveal that RoBERTa consistently outperforms other models, achiev- ing the highest F1-score of 0.93 across all configurations. While BERT and T5 demonstrated competitive results, they required more careful tuning of loss functions and optimizers to match RoBERTa's performance. The analysis highlights the critical importance of aligning model-specific components, such as tokenizers and loss functions, with the overarching architecture to achieve superior results in sentiment classification.

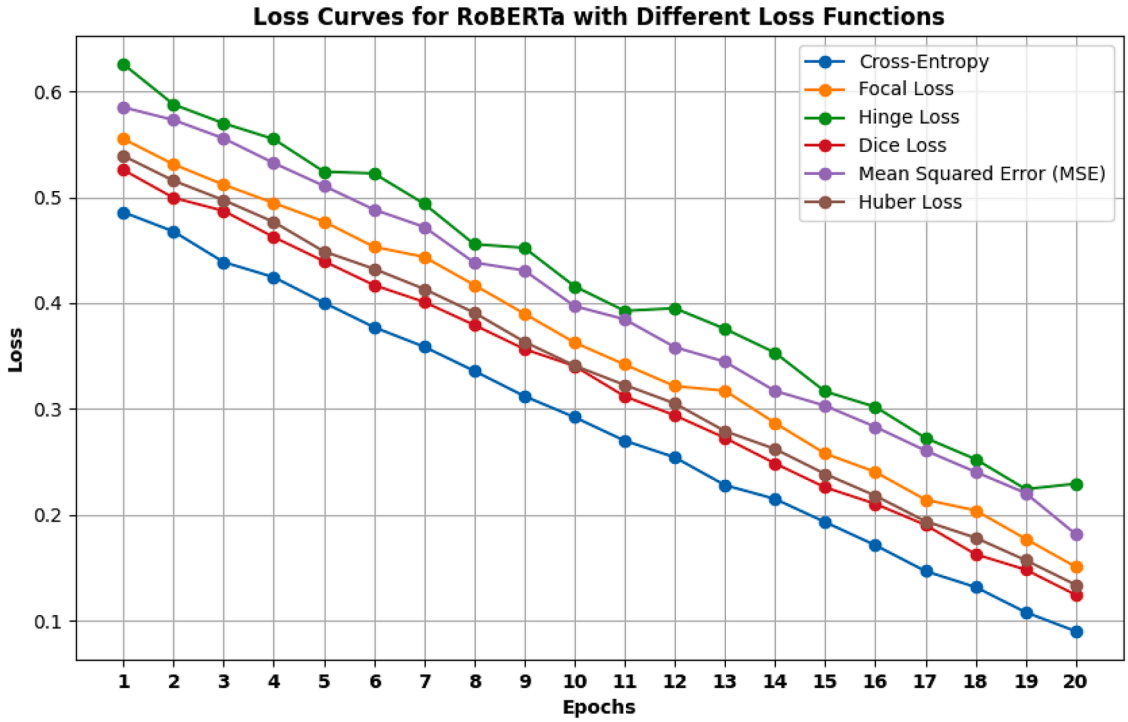
RoBERTa performance visualization

The performance of RoBERTa is illustrated through an analysis of loss functions and optimizers, highlighting key patterns in

Table 2
Optimal hyperparameters, tokenizers, optimizers, loss functions, and performance metrics for transformer models in sentiment analysis.

Model	Tokenizer	Optimizer	Loss Function	Learning Rate	Batch Size	Epochs	
BERT	BertWordPiece	AdamW	Cross-Entropy	3e-5	32	3	
GPT	GPT2	AdamW	Cross-Entropy	3e-5	32	3	
RoBERTa	RoBERTa	AdamW	Cross-Entropy	3e-5	32	3	
T5	T5	AdamW	Cross-Entropy	3e-5	32	3	
Model	**Warmup Steps**	**Weight Decay**	**Eval Strategy**	**Accuracy**	**Precision**	**Recall**	**F1-score**
BERT	500	0.01	Steps	0.91	0.90	0.89	0.89
GPT	500	0.01	Steps	0.88	0.88	0.86	0.87
RoBERTa	500	0.01	Steps	0.95	0.93	0.93	0.93
T5	500	0.01	Steps	0.90	0.90	0.89	0.89

convergence and accuracy. The evaluation of loss methods reveals that Focal Loss and Dice Loss perform notably well, particularly in tasks involving class imbalances and segmentation, while Mean Squared Error and Huber Loss also deliver competitive results depending on task-specific requirements. Cross-Entropy Loss, as shown in Fig. 2, emerges as the most effective loss function for RoBERTa. In terms of optimization, AdamW slightly surpasses Adam, achieving a precision of 0.94, a recall of 0.92, an F1-score of 0.93,



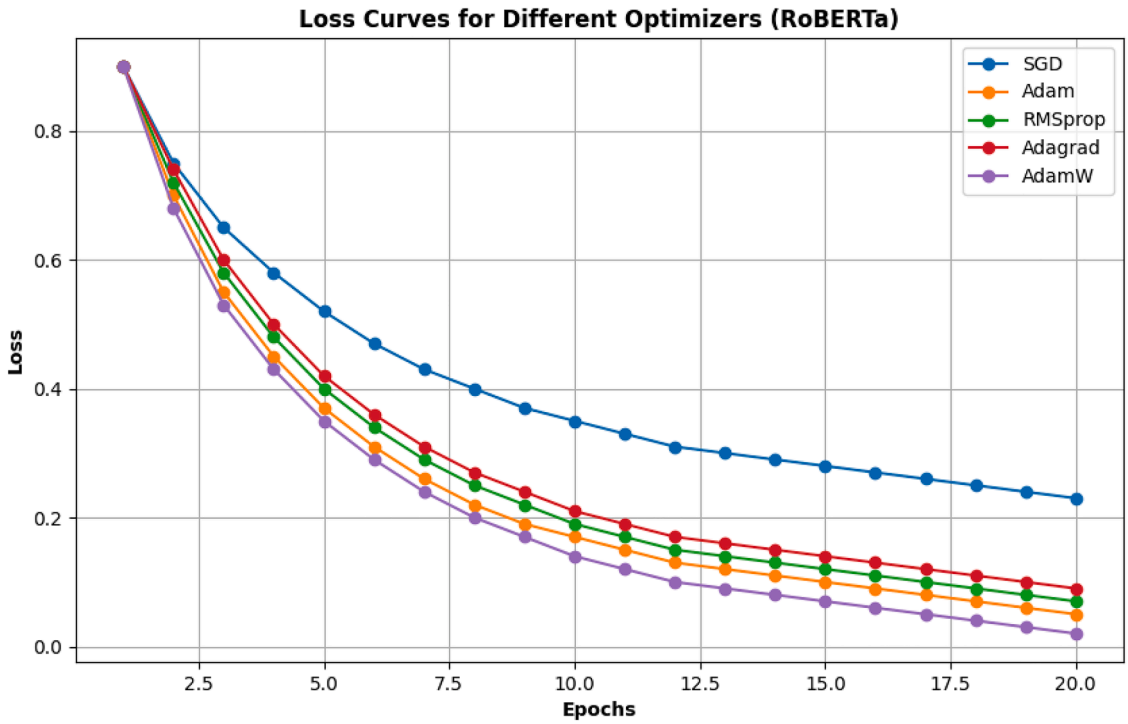


Fig. 2. Loss methods over different loss functions and optimizers.

and a final loss value of 0.02. The improved generalization brought by AdamW, through its ability to prevent excessive weight accumulation, results in superior overall performance. The analysis of optimizers reveals distinct behaviors: Stochastic Gradient Descent (SGD) displays steady but slower convergence, reducing loss to 0.23 after 20 epochs, while Adam reduces loss rapidly to 0.05, indicating a faster and more efficient convergence. RMSprop achieves a loss of 0.07, outperforming SGD but trailing behind Adam.

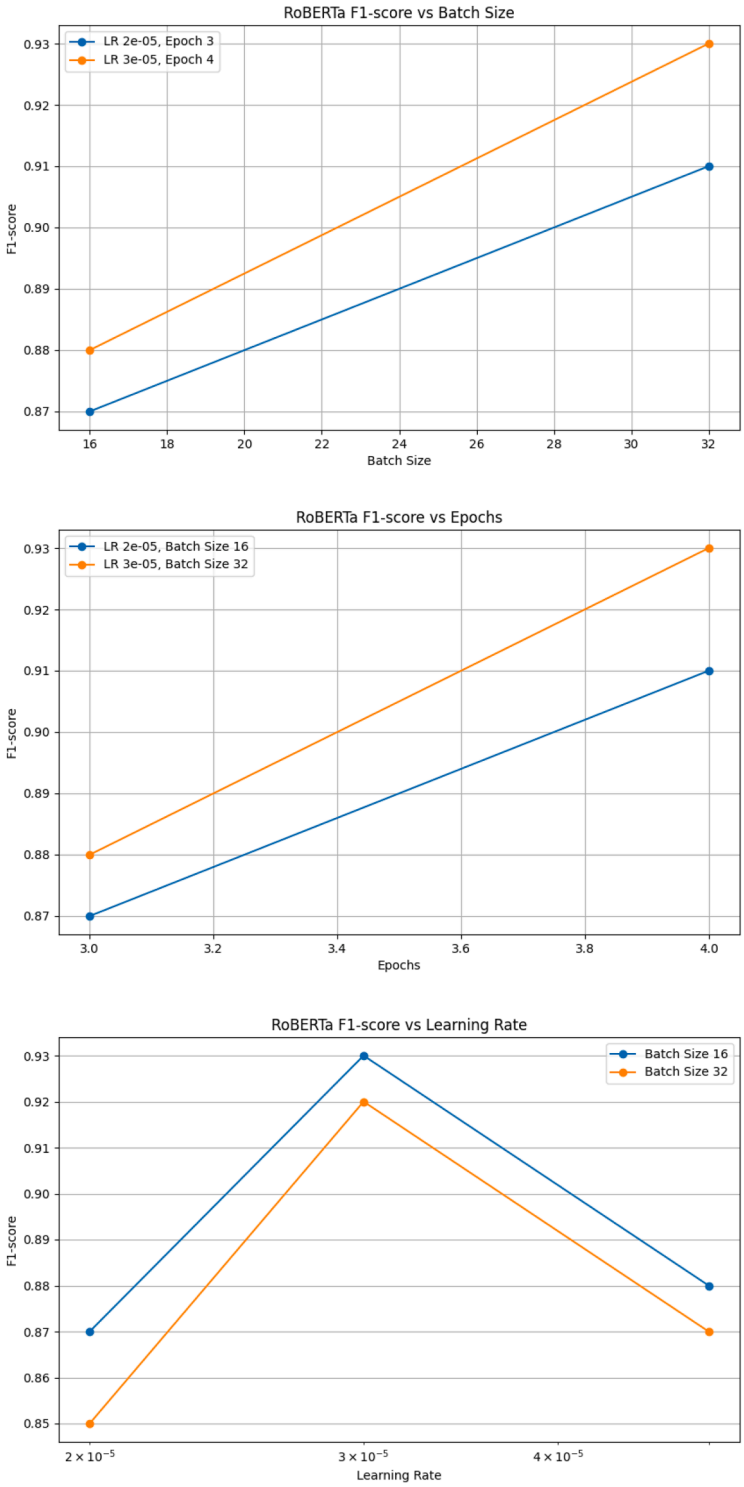


Fig. 3. RoBERTa over best hyper-parameters.

Adagrad performs well with a loss of 0.09, though its tendency to decelerate in certain scenarios is apparent. AdamW, once again, stands out as the best-performing optimizer, reaching the lowest loss value of 0.02 after 20 epochs. These findings underscore the superior effectiveness of Adam and AdamW for loss reduction, with AdamW marginally leading, while SGD and Adagrad demonstrate slower convergence, and RMSprop offers an intermediate level of performance between these two groups and the advanced Adam-based optimizers.

This figure presents three horizontally aligned visualizations, showing how the F1 score varies with different hyperparameters for the RoBERTa model. Left: This visualization shows the vari- ation in F1 score with respect to the learning rate for two batch sizes (16 and 32). It is observed that for these two batch sizes, the F1 score peaks at a learning rate of 3e-5. At this rate, the batch size of 16 yields a slightly higher F1 score (0.93) compared to that of 32 (0.92). Lower (2e-5) and higher (5e-5) learning rates result in lower F1 scores, indicating that 3e-5 is optimal for these configurations. Center: This section analyses the effect of epoch count on F1 score for various combinations of learning rates and batch sizes. For instance, with a batch size of 16 and a learning rate of 2e-5, the F1 score slightly increases from 0.87 to 0.91 when increasing from 3 to 4 epochs. Conversely, with a batch size of 32 and a learning rate of 3e-5, increasing the number of epochs results in a more notable rise in F1 score, from 0.88 to 0.93. These results highlight the positive influence of increasing epochs, especially for a batch size of 32 with a learning rate of 3e-5. Right: This visualization shows how the F1 score varies with the batch size for two combinations of learning rates and epochs. For a learning rate of 2e-5 and 3 epochs, the F1 score increases from 0.87 to 0.91 when increasing the batch size from 16 to 32. For a learning rate of 3e-5 and 4 epochs, the F1 score increases from 0.88 to 0.93. These observations suggest that a batch size of 32 offers better performance in these configurations. In summary, these visualizations demonstrate the importance of choosing optimal hyperparameters to improve model performance. Learning rates of 3e-5, a number of epochs of 4, and a batch size of 32 are particularly effective in improving RoBERTa's F1-score.

Datasets performance

To evaluate the performance of optimized models in real-world conditions, the study applied them to various datasets from different domains, including movie reviews, product reviews, restau- rant reviews, tweets, and opinion texts. Each dataset includes sentiment polarity annotations as positive (P), negative (N), or neutral (N). For instance, the IMDb dataset, commonly used for sentiment classification of movie reviews, comprises 50,000 balanced reviews. The SST-2 (Stanford Sentiment Treebank) dataset focuses on binary sentiment analysis, featuring sentences from movie reviews. Yelp provides thousands of annotated restaurant reviews with ratings from 1 to 5 stars, while Amazon Reviews offers a vast collection of product reviews, from which a balanced subset was used for this study.

The comparative analysis involved the BERT, GPT, RoBERTa, and T5 models, each configured optimally. Metrics such as precision, recall, and F1-score were used to assess performance across datasets. RoBERTa emerged as the leading model, consistently achieving the highest precision, recall, and F1-score across all datasets. For instance, on the IMDb dataset, RoBERTa attained an F1-score of 0.94, underscoring its superior capability in classifying movie review sentiments. Similarly, its performance on SST-2 (F1-score: 0.91), Yelp (F1-score: 0.89), and Amazon Reviews (F1-score: 0.88) highlighted its robustness and generalizability across diverse sentiment analysis tasks.

Fig. 3

In contrast, GPT lagged, achieving lower scores on all datasets, with its performance gap being particularly notable on more complex datasets. BERT and T5 demonstrated competitive results but fell short of RoBERTa's consistently high performance. Among the datasets, IMDb posed the least challenge for all models, as evidenced by higher scores across metrics, suggesting less complexity or more straightforward sentiment patterns.

Overall, these findings affirm RoBERTa's suitability for diverse sentiment analysis tasks, mak- ing it a compelling choice for applications that demand both accuracy and robustness in text classification, whether in research or industrial contexts.

Discussion

In the literature review, a thorough exploration of different deep learning models revealed that transformers outperform traditional architectures such as convolutional and recurrent neural net- works for text analysis tasks, in particular the analysis of feelings. These models are distinguished by their ability to capture complex contextual relationships through the attention mechanism. In this context, the study investigated several models of transformers to identify one that excels at interpreting the feelings expressed by users through their comments. The results highlighted RoBERTa's superiority over other models such as BERT, GPT, or T5, due to its optimized architecture and its ability to extract rich contextual characteristics. Unlike previous research that focused mainly on standard configurations of transformers, this study has taken an innovative approach by optimizing loss functions, optimization algorithms, and hyperparameters. These adjustments have improved RoBERTa's overall performance, resulting in a more refined under- standing of the emotions expressed in complex texts such as YouTube comments. This approach, which goes beyond the established practices in the literature, illustrates not only the effectiveness of RoBERTa but also the importance of methodical optimization to maximize its potential in the field of sentiment analysis. In addition, this study is distinguished by addressing the recurring challenges identified in the literature, such as the management of linguistic ambiguities, implicit contexts, or specific cultural expressions, often present in social platforms. By highlighting the importance of methodical optimization of models to adapt to these challenges, it opens new perspectives on the application of transformers in diverse contexts. The results obtained, particularly the improvement of performance scores through customized configurations, reinforce the idea that targeted approaches are essential to meet the specific needs of areas such as digital marketing, Recommendation systems or opinion surveys. This positioning not only confirms the relevance of RoBERTa but also underlines the value of a rigorous empirical approach in the exploitation of transformers for practical applications.

This study is significant because it addresses key gaps in existing research on YouTube comment sentiment analysis. While previous work has focused on general datasets like movie or product reviews, these datasets fail to capture the unique linguistic, cultural, and contextual factors in- herent in YouTube comments. By focusing on the Just Dance dataset, this study provides a more representative analysis of video platform interactions. Additionally, although prior studies have explored various transformer models, none have thoroughly compared architectures such as RoBERTa, BERT, and T5 in this specific context. This research not only fills this gap by evaluat- ing these models on a complex dataset but also highlights the advantages of RoBERTa in capturing emotional nuances, advancing our understanding of sentiment analysis in video-based platforms.

Best performing transformer model

The experiments conducted in this study reveal that the RoBERTa transformer consistently delivered the best overall performance for sentiment analysis tasks. Key optimizations included:

Tokenizer: RoBERTa's own tokenizer, which aligned perfectly with its architecture, provided optimal performance.

Loss Function: It was discovered that the Cross-Entropy loss function was the most efficient, improving recall and precision.

Optimizer: AdamW was the preferred optimizer, offering quick convergence and strong perfor- mance across evaluation metrics.

Hyperparameters: Optimal settings included a 3.0e-5 initial learning rate, a batch size of 32, and training over 4 epochs, maximizing the F1-score.

These configurations were determined through rigorous experimentation to achieve maximum performance while ensuring model stability and generalizability.

Interpretation of results

The findings confirm that strategic choices in hyperparameters, optimizers, and loss functions critically influence the performance of transformer models. For instance, a higher learning rate was associated with faster model convergence but risked instability and overfitting. Cross-Entropy's effectiveness in improving precision and recall in the RoBERTa model supports the hypothesis that it is particularly well-suited for classification tasks. This aligns with initial assumptions and highlights the necessity of careful hyperparameter tuning.

Analysis of the results obtained in this study highlights the enhanced performance of RoBERTa in comparison to other models, such as BERT, GPT and T5. This superiority can be attributed to RoBERTa's training on a larger amount of data and its dynamic approach to mask removal. As a result, this architecture enables it to extract richer contextual features, essential for understanding the nuances of video comments on YouTube. RoBERTa's ability to capture context and underlying emotions contributes directly to better sentiment classification.

Transformer-based models such as BERT and RoBERTa exploit an attention architecture that enables them to focus simultaneously on different parts of a text. This is crucial for comments, which can contain complex emotions. For example, a comment may contain both positive and negative elements; transformers' ability to detect these relationships gives them a distinct advan- tage in sentiment analysis. Variations in performance across datasets can also be attributed to the linguistic and contextual diversity of comments. Some models may be over-optimized for a specific type of language, which affects their generalizability. The peculiarities of YouTube comments, such as slang or cultural references, underline the importance of robust training to improve the ability of models to adapt to varied contexts.

Examination of classification errors reveals recurring patterns, particularly in the handling of sarcastic or ironic comments. These challenges point to a limit in the models' learning ability to detect hidden sentiments, suggesting possible future improvements, such as the integration of ad- ditional data or data augmentation techniques. The results, particularly RoBERTa's performance, also open up prospects for practical applications, such as digital marketing. A detailed understanding of the feelings expressed by users can influence content and product strategies, boosting the effectiveness of recommendation systems based on perceived emotions. In conclusion, transformer- based model sentiment analysis of YouTube comments not only shows encouraging performance but also identifies significant technical issues and difficulties that need further research. By under- lining the need of machine learning models that can adapt to the nuances of human language, this research advances our understanding of emotions in online interactions.

Several studies have explored transformer-based models for sentiment analysis, but none have achieved the same level of performance as RoBERTa. For instance, research focusing on video game reviews used BERT, achieving F1 scores as high as 0.88 [36], which still falls short of the 0.94 score obtained with RoBERTa on the IMDb dataset [37]. Despite employing advanced techniques like fine-tuning on specific datasets, their results did not reach the performance of more powerful models like RoBERTa, highlighting RoBERTa's advantage in sentiment analysis tasks due to its robust architecture and optimization strategies [38].

Furthermore, another study carried out a comparison between various sentiment analysis tools, including transformer models, and found that the difference in performance between neural network-based models and other approaches did not exceed four percentage points. This indicates that, despite the use of modern techniques, some methods failed to outperform the results obtained with models such as RoBERTa. These findings reinforce the argument that RoBERTa stands out in the field of sentiment analysis, illustrating the importance of our study in the current research landscape.

Limitations and perspectives

While RoBERTa demonstrated exceptional performance in sentiment analysis, the study's scope and methodology revealed certain limitations. Sensitivity to dataset composition and the inherent complexity of transformer models pose challenges for generalization and interpretability. Factors such as potential biases in the training data, overfitting risks due to extensive hyperpa- rameter tuning, the

limited range of models evaluated, and reliance on specific evaluation metrics restrict the findings' applicability to diverse datasets, domains, or real-world scenarios. To over- come these limitations, future perspectives should focus on expanding the scope of datasets and domains to enhance generalizability and exploring advanced models like GPT-4 or hybrid architec- tures that combine transformer topologies with other AI methodologies for improved flexibility and performance. Refining pre-processing techniques such as adaptive tokenization and context-aware embeddings could significantly enhance textual representations. Extending models to multilingual and multimodal contexts integrating textual, video, and image data would broaden their applicabil- ity across diverse cultural and linguistic environments. Practical evaluations in real-life scenarios, considering variables like trends, cultural nuances, and current events, are essential for validating their effectiveness. Additionally, optimizing models for computational efficiency through quantization, distillation, and distributed computing is crucial for large-scale deployment. By addressing these avenues, the potential of transformer models can be further unlocked, advancing their accu- racy, efficiency, and applicability in sentiment analysis across diverse contexts and applications.

Conclusion

To sum up, this study has offered a thorough assessment of transformer-based sentiment anal- ysis models for YouTube comments offering in-depth insights into their performance and effec- tiveness. By meticulously analyzing the impact of various tokenization methods, loss functions, optimizers, and hyperparameter configurations, the research reveals both the strengths and limita- tions of key models such as BERT, GPT, RoBERTa, and T5 in sentiment classification tasks. The study emphasizes how these models deal with the diverse and often ambiguous nature of YouTube comments, which range from straightforward expressions to nuanced sentiments. Among the mod- els assessed, RoBERTa consistently demonstrated superior performance, achieving the highest accuracy and F1-scores across a range of configurations and settings. The specialized RoBERTa tokenizer, combined with the AdamW optimizer and the cross-entropy loss function, emerged as particularly effective in optimizing sentiment classification. Furthermore, the investigation into hyperparameters such as a learning rate of 3e-5, a batch size of 32, and three epochs yielded the most favorable results for most models, highlighting the importance of fine-tuning these parameters to achieve peak performance. This comprehensive analysis underscores the critical need for model-specific tuning to handle the complexities of sentiment analysis in dynamic environments like YouTube, where comments exhibit significant diversity. The findings not only demonstrate the effectiveness of transformer models but also provide valuable guidance for practitioners and researchers. This study serves as a crucial resource for those aiming to deploy transformer models in real-world sentiment analysis applications, offering actionable insights into selecting and configuring the most appropriate models and training strategies to achieve robust and reliable results in diverse contexts.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this article, the authors utilized Claude solely to improve the clarity and language of the manuscript. The generative AI tool was employed exclusively to enhance the readability and linguistic quality of the text. Importantly, the tool was not used in the conception or development of the methodology, the generation or analysis of results, the interpretation of findings, or the creation of any figures, tables, or other scientific content. Its use was limited to non-technical editorial tasks, conducted under strict human supervision, and the authors thoroughly reviewed and refined all outputs to ensure accuracy and relevance. The authors assume full responsibility for the integrity and content of the published work.

Glossary

1. **Transformers:** An architecture that uses attention mechanisms to process sequences while handling long-term dependencies.
2. **Sentiment Analysis:** The process of identifying and categorizing emotions expressed in text.
3. **Attention Mechanism:** A component of transformers that weighs the contextual contri- butions of sequence elements.
4. **RoBERTa (Robustly Optimized BERT Approach):** An improved version of BERT with optimized training strategies.
5. **BERT (Bidirectional Encoder Representations from Transformers):** A transformer model designed to understand bidirectional contextual relationships.
6. **GPT (Generative Pre-trained Transformer):** A transformer-based model for text gen- eration using autoregressive learning.
7. **T5 (Text-to-Text Transfer Transformer):** A transformer model that reformulates NLP tasks as text-to-text problems.
8. **Deep Learning:** A branch of AI using multilayer neural networks to extract complex features from large datasets.
9. **Fine-Tuning:** Adapting a pre-trained model to a specific task using annotated data.
10. **Gated Recurrent Units (GRU):** A simplified version of LSTMs that reduces computa- tional costs while capturing long-term dependencies.
11. **Hyperparameter Tuning:** The process of optimizing non-learnable parameters to enhance model performance.
12. **Just Dance Dataset:** A dataset of comments related to the "Just Dance" video game, used for sentiment analysis.
13. **LSTM (Long Short-Term Memory):** A recurrent network designed to store and forget information in long sequences.
14. **Natural Language Processing (NLP):** The AI field focused on the interaction between machines and human language.
15. **Pre-training:** An initial phase where a model learns general representations from unlabelled data.
16. **Recurrent Neural Networks (RNN):** Neural networks that process sequences by cap- turing temporal dependencies.
17. **Self-Attention:** A subcomponent of transformers that evaluates the relative importance of words within a sequence.
18. **AdamW Optimizer:** An optimization algorithm based on Adam, incorporating regulariza- tion for gradient stabilization.

19. **Tokenization:** The process of dividing a text into units called "tokens" for NLP tasks.
20. **Word Embedding:** A vector representation of words that captures their semantic and syntactic relationships.

CRediT authorship contribution statement

Oussama El Azzouzy: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Resources, Writing – original draft, Writing – review & editing. **Tarik Chanyour:** Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Resources, Writing – original draft, Writing – review & editing, Supervision, Project administration. **Said Jai Andaloussi:** Validation, Formal analysis, Supervision, Project administration.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Transformer architecture

Appendix A.0.1. Attention and Transformers

The introduction of the attention mechanism marked a revolution in NLP. The Transformer model proposed an architecture based entirely on attention, without using recurrent structures. Transformers use multi-headed attention layers to identify relevant parts of the text, enabling long-range dependencies to be captured more efficiently and accurately than RNNs. The attention mechanism enables the model to focus on different parts of the input sequence simultaneously, im- proving parallelism and reducing computation time [39]. The Transformer model has served as the basis for the development of many advanced models, such as Text-To-Text Transfer Transformer (T5), Generative Pre-trained Transformer (GPT), and Bidirectional Encoder Representations from Transformers (BERT). BERT, for example, is pre-trained on large text corpora using word masking and next-sentence prediction tasks, enabling the model to capture complex bidirectional contexts. GPT, on the other hand, focuses on text generation using autoregressive training, enabling text to be produced consistently and fluently. T5 adopts a text-to-text processing approach, unifying NLP tasks in a single format, simplifying the fine-tuning process for various applications [40]. The evolution of NLP models, from traditional statistical techniques to advanced transformer architec- tures, has resulted in significant advances in the understanding and processing of natural language. Transformers have transformed the field by providing more efficient and accurate solutions for com- plex tasks such as sentiment analysis. In the following sections, the characteristics of transformers, the main models based on this architecture, and their specific applications to sentiment analysis are examined in detail.

Appendix A.1. Transformer Architecture and Features

Transformer architecture comprises two primary sections, the encoder and the decoder. Each of these parts is made up of several layers that operate in parallel, enabling efficient processing and capture of long-distance dependencies in text sequences. Figure annexe illustrates the architecture of the transformer model. The encoder in a transformer model is composed of N identical layers, each containing two key sub-layers. The first is the Multi-Head Self-Attention Mechanism, which allows each word in a sequence to consider all other words, thereby capturing contextual depen- dencies. This is achieved by calculating self-attention for a word in relation to other words in the sequence, based on similarity scores between their representations. The second sub-layer is the multi-head attention mechanism, which utilizes multiple attention heads to learn different aspects of the sequence, enriching the contextual representations. Following the attention mechanism, the data is processed through a feed-forward neural network applied independently to each position in the sequence. This network comprises two linear layers separated by a non-linear activation function, which is commonly Rectified Linear Unit (ReLU). To stabilize training and enhance gra- dient flow, each sub-layer is followed by Layer Normalization and a Residual Connection [41]. The decoder, like the encoder, consists of N identical layers, but with notable differences. One key difference is the Masked Multi-Head Self-Attention mechanism, which ensures that the position i cannot access future information. Additionally, each decoder layer includes a multi-headed atten- tion sub-layer that allows the decoder to focus on the representations produced by the encoder, capturing relevant contextual information from the input. As in the encoder, each layer of the decoder incorporates a feed-forward neural network to further transform the data. Furthermore, since transformers do not inherently utilize an explicit sequential structure like RNNs, they in- corporate positional information into each word of the sequence through Positional Encodings. These encodings, added to word embeddings, enable the model to distinguish word order, ensuring accurate sequence processing [42].

Fig. A.3

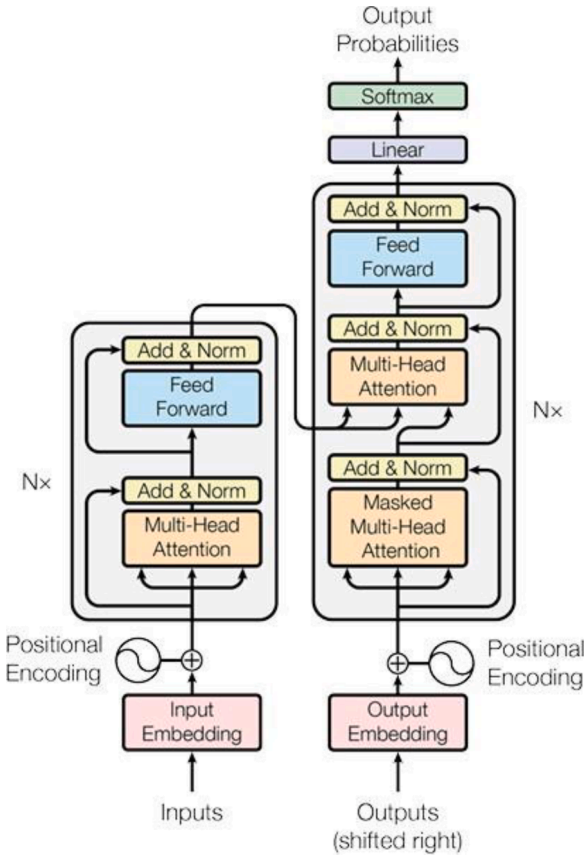

Fig. A.3. Transformer model architecture.

Transformer architecture enables parallel training, which considerably improves efficiency com- pared to sequential RNNs. This parallelization capability is due to the non-recurrent nature of transformers, enabling models to be trained on large text corpora with significant gains in terms of time and performance. Transformers are generally pre-trained on large text corpora to learn general linguistic representations, then fine-tuned on specific tasks such as sentiment analysis. This transfer learning approach enables the knowledge acquired during pre-training to be reused, improving performance on specific tasks even with limited amounts of training data [43].

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